

1)Converting binary to hexadecimal:

a) 1110

3rd bit weight = $2^3 = 8$ 2nd = $2^2 = 4$ 1st = $2^1 = 2$ 0th = $2^0 = 1$ VALUE = $(1 \times 8) + (1 \times 4) + (1 \times 2) + (0 \times 1) = 14 = \mathbf{E \text{ or } E_16}$

b) 10

Write as 0010

 $0+0+2+0 = 2$ or **2_16**

c)10111

Right group = 0111

 $(0 \times 8) + (1 \times 4) + (1 \times 2) + (1 \times 1) = 7$ Then combine 1 and 7 = **17** or **17_16**

Left group = 0001

 $(0 \times 8) + (0 \times 4) + (0 \times 2) + (1 \times 1) = 1$

d)10100110

Split into 4 bit groups

Right group = 0110

 $8+2 = 10 = \mathbf{A}$ Then combine **A6** or **A_16**

Left group = 1010

 $4+2 = 6$

e)1111110000

Right group = 0000

0

Combined, **3F0**

Middle = 1111

 $8+4+2+1 = 15 = \mathbf{F}$

Left = 11 (0011)- pad with zeros

 $2+1 = 3$

f)100110000010

Right group = 0010

2

Combined, **982** or **982_16**

Middle = 1000

8

Left = 1001

 $8+1 = 9$

2) Converting hexadecimal to decimal

a) 23 (base 16)

3 has weight 1, 2 has weight 16

$$2 \times 16 = 32$$

$$3 \times 1 = 3 \quad 32 + 3 = \mathbf{35} \text{ or } \mathbf{35_{10}}$$

b) 92 (base 16)

$$9 \times 16 = 144, \quad 2 \times 1 = 2 \quad \text{-----} \rightarrow 144 + 2 = \mathbf{146} \text{ or } \mathbf{146_{10}}$$

c) 1A (base 16)

A = 10, has weight 1 and 1 has weight 16

$$10 \times 1 = 10, \quad 1 \times 16 = 16 \quad \text{-----} \rightarrow 10 + 16 = \mathbf{26} \text{ or } \mathbf{26_{10}}$$

d) 8D (base 16)

D = 13, weight = 1, 8 has weight 16

$$13 \times 1 = 13, \quad 8 \times 16 = 128 \quad \text{-----} \rightarrow 128 + 13 = \mathbf{141} \text{ or } \mathbf{141_{10}}$$

e) F3 (base 16)

$$3 \times 1 = 3, \quad F = 15, 15 \times 16 = 240 \quad \text{-----} \rightarrow 3 + 240 = \mathbf{243} \text{ or } \mathbf{243_{10}}$$

f) EB (base 16)

$$11 \times 1 = 11, \quad 14 \times 16 = 224 \quad \text{-----} \rightarrow 11 + 224 = \mathbf{235} \text{ or } \mathbf{235_{10}}$$

g) 5C2 (base 16)

$$5 \times 256 = 1280,$$

$$12 \times 16 = 192,$$

$$2 \times 1 = 2 \quad \text{-----} \rightarrow 1280 + 192 + 2 = \mathbf{1474} \text{ or } \mathbf{1474_{10}}$$

h) 700 (base 16)

$$(7 \times 256) + (0 \times 16) + (0 \times 1) = \mathbf{1792} \text{ or } \mathbf{1792_{10}}$$

3) Converting each binary number to Gray code

*MSB (Left most bit) of the Gray code is always the same as the MSB of the binary number

a) 11011

Bit 5 (MSB) = 1, Bit 4 = 1, Bit 3 = 0, Bit 2 = 1, Bit 1 (LSB) = 1

Now we determine each gray code bit step by step:

MSB = 1

*1st gray bit (MSB) = 1

*2nd gray bit = (bit 5 + bit 4) $\rightarrow 1 + 1 = 0$ (ignore carry)

*3rd gray bit = (bit 4 + bit 3) $\rightarrow 1 + 0 = 1$

*4th gray bit = (bit 3 + bit 2) $\rightarrow 0 + 1 = 1$

*5th gray bit = (bit 2 + bit 1) $\rightarrow 1 + 1 = 0$ (ignore carry)

Combine the gray code bits together from left to right:

10110

b) 1001010

Bit 7 (MSB) = 1, Bit 6 = 0, Bit 5 = 0, Bit 4 = 1, Bit 3 (LSB) = 0, Bit 2 = 1, Bit 1 (LSB) = 0

Now determine each gray code step by step:

*1st gray bit (MSB) = 1

*2nd gray bit = (bit 7 + 6) $\rightarrow 1 + 0 = 1$

*3rd gray bit = (bit 6 + 5) $\rightarrow 0 + 0 = 0$

*4th gray bit = (bit 5 + 4) $\rightarrow 0 + 1 = 1$

*5th gray bit = (bit 4 + 3) $\rightarrow 1 + 0 = 1$

*6th gray bit = (bit 3 + 2) $\rightarrow 0 + 1 = 1$

*7th gray bit = (bit 2 + 1) $\rightarrow 1 + 0 = 1$

Combine the gray code bits together from left to right:

1101111

c) 1111011101110

B13 = 1, B12 = 1, B11 = 1, B10 = 1, B9 = 0, B8 = 1, B7 = 1, B6 = 1, B5 = 0, B4 = 1, B3 = 1, B2 = 1, B1 = 0

Gray codes:

* 1st gray bit (MSB) = 1

*2nd gray bit = (bit 13 + bit 12) = 1+1=0(ignore carry)
 *3rd gray bit = (bit 12 + bit 11) = 1+1=0
 *4th gray bit = (bit11 + bit 10) = 1+1=0
 *5th gray bit = (bit 10 +bit 9) = 1+0 = 1
 *6th gray bit = (bit 9 + bit 8) = 0 + 1 = 1
 *7th gray bit = (bit 8 + bit 7) = 1 + 1 = 0
 *8th gray bit = (bit 7 + bit 6) = 1 + 1 = 0
 *9th gray bit = (bit 6 +bit5) = 1 + 0 = 1
 *10th gray bit = (bit 5+bit 4) = 0 + 1 = 1
 *11th gray bit = (bit 4 + bit 3) = 1 + 1 = 0
 *12th gray bit = (bit 3+ bit 2) = 1 + 1 = 0
 *13th gray bit = (bit 2 + 1) = 1 + 0 = 1

Combine the gray code bits together from left to right:
1000110011001

4) Converting gray code to binary

a) 1010

G4(MSB) = 1, G3 = 0, G2 = 1, G1(LSB) = 0

Determining each binary bit:

*1st binary bit(B4, MSB) : keep the leftmost gray bit ----> B4 = 1
 *2nd binary bit(B3) = current binary bit(B4) + Next gray bit(G3) = 1 + 0 = 1
 *3rd binary bit(B2) = current binary bit(B3) + Next gray bit(G2) = 1 + 1 = 0(ignore carry)
 4th binary bit(B1) = current binary bit(B2) + Next gray bit(G1) = 0 + 0 = 0

Combining binary bits from left to right:
1100

b) 00010

G5(MSB) = 0, G4 = 0, G3 = 0, G2 = 1, G1(LSB) = 0

Determining each binary bit:

*1st binary bit(B5, MSB) = 0
 *2nd binary bit(B4) = current binary bit(B5) + Next gray bit(G4) = 0 + 0 = 0
 *3rd binary bit(B3) = current binary bit(B4) + Next gray bit(G3) = 0 + 0 = 0
 *4th binary bit(B2) = current binary bit(B3) + Next gray bit(G2) = 0 + 1 = 1
 *5th binary bit(B1) = current binary bit(B2) + Next gray bit(G1) = 1 + 0 = 1

Combining binary bits from left to right:
00011

c) 11000010001

G11(MSB) = 1, G10 = 1, G9 = 0, G8 = 0, G7 = 0, G6 = 0, G5 = 1, G4 = 0, G3 = 0, G2 = 0, G1(LSB)

Determining each binary bit:

*1st binary bit(B11, MSB) = 1

*2nd binary bit(B10) = current binary bit(B11) + Next gray bit(G10) = 1 + 1 = 0

*3rd binary bit(B9) = current binary bit(B10) + Next gray bit(G9) = 0 + 0 = 0

*4th binary bit(B8) = current binary bit(B9) + Next gray bit(G8) = 0 + 0 = 0

*5th binary bit(B7) = current binary bit(B8) + Next gray bit(G7) = 0 + 0 = 0

*6th binary bit(B6) = current binary bit(B7) + Next gray bit(G6) = 0 + 0 = 0

*7th binary bit(B5) = current binary bit(B6) + Next gray bit(G5) = 0 + 1 = 1

*8th binary bit(B4) = current binary bit(B5) + Next gray bit(G4) = 1 + 0 = 1

*9th binary bit(B3) = current binary bit(B4) + Next gray bit(G3) = 1 + 0 = 1

*10th binary bit(B2) = current binary bit(B3) + Next gray bit(G2) = 1 + 0 = 1

*11th binary bit(B1) = current binary bit(B2) + Next gray bit(G1) = 1 + 1 = 0

Combining binary bits from left to right:

10000011110

CHAPTER 3

5) Repeat problem 5 for the waveforms in the figure(shown in Problem Set)

Interval 1(t1):

Input A is HIGH(1), Input B is LOW(0), Therefore output X = LOW(0)

Interval 2(t2):

Input A is LOW(0), Input B is HIGH(1), Therefore output X = LOW(0)

Interval 3 (t3):

Input A is HIGH(1), Input B is LOW(0), Therefore output X = LOW(0)

Interval 4 (t4):

Input A is LOW(0), Input B is HIGH(1), Therefore output X = LOW(0)

Interval 5 (t5):

Input A is HIGH(1), Input B is LOW(0), Therefore output X = LOW(0)

Interval 6 (t6):

Input A is LOW(1), Input B is HIGH(0), Therefore output X = LOW(0)

Since A and B are completely out of phase, output stays LOW(0)

Output X waveform representation

HIGH(1) |

LOW(0) | _____

6) Output waveform for 3 input AND gate

Logic Rule for a 3-input AND Gate:

- Output X is HIGH(1) only when all three inputs are HIGH(1) at the same time.
- If any input is LOW(0), Output X is LOW(0)

Interval 1(t1):

*Input A = HIGH(1)

*Input B = LOW(0)

*Input C = HIGH(1)

*Not all inputs are HIGH(B is LOW), Therefore, Output X = LOW(0)

Interval 2(t2):

*Input A = LOW(0)

*Input B = HIGH(1)

*Input C = LOW(0)

*Not all inputs are HIGH, Therefore, Output X = LOW(0)

Interval 3(t3):

*Input A = HIGH(1)

*Input B = HIGH(1)

*Input C = HIGH(1)

*All three inputs are HIGH, Therefore, Output X = HIGH(1)

Interval 4(t4):

*Input A = LOW(0)

*Input B = LOW(0)

*Input C = LOW(0)

*All inputs are LOW, Therefore, Output X = LOW(0)

Interval 5(t5):

*Input A = HIGH(1)

*Input B = HIGH(1)

*Input C = HIGH(1)

*All inputs are HIGH, Therefore, Output X = HIGH(1)

Interval 6(t6):

*Input A = LOW(0)

*Input B = HIGH(1)

*Input C = LOW(0)

*Not all inputs are HIGH, Therefore, Output X = LOW(0)

Timing Analysis Table

Interval | Input A | Input B | Input C | Output X

t1	1	0	1	0 (LOW)
t2	0	1	0	0 (LOW)
t3	1	1	1	1 (HIGH)
t4	0	0	0	0 (LOW)
t5	1	1	1	1 (HIGH)
t6	0	1	0	0 (LOW)

7) Problem for a 4 input OR gate.

Logic Rule for a 3-input AND Gate:

- Output X is HIGH(1) if at least one input is HIGH(1).
- Output is LOW(0) only when all inputs are LOW(0).

Interval 1(t1):

*Input A = HIGH(1), *Input B = LOW(0)

*Input C = HIGH(1), *Input D = LOW(0)

Output X = HIGH(1)

Interval 2(t2):

*Input A = LOW(0), *Input B = HIGH(1)

*Input C = LOW(0), *Input D = LOW(0)

Output X = HIGH(1)

Interval 3(t3):

*Input A = HIGH(1), *Input B = LOW(0)

*Input C = HIGH(1), *Input D = LOW(0)

Output X = HIGH(1)

Interval 4(t4):

*Input A = LOW(0), *Input B = HIGH(1)

*Input C = LOW(0), *Input D = HIGH(1)

Output X = HIGH(1)

Interval 5(t5):

*Input A = HIGH(1), *Input B = LOW(0)

*Input C = HIGH(1), *Input D = HIGH(1)

Output X = HIGH(1)

Interval 6(t6):

*Input A = LOW(0), *Input B = HIGH(1)

*Input C = LOW(0), *Input D = HIGH(1)

Output X = HIGH(1)

Timing Analysis Table

Interval | Input A | Input B | Input C | Input D | Output X

t1		1		0		1		0		1 (HIGH)
t2		0		1		0		0		1 (HIGH)
t3		1		0		1		0		1 (HIGH)
t4		0		1		0		1		1 (HIGH)
t5		1		0		1		1		1 (HIGH)
t6		0		1		0		1		1 (HIGH)

8) Output waveform for a NAND gate

Gate type & logic rule for NAND gate

- The output X is LOW(0) only when both inputs (A and B) are HIGH(1)
- If either input (or both) is LOW(0), the output X is HIGH(1).

First Half (input B is LOW):

*Input B = LOW(0) throughout

*Input A goes between HIGH and LOW

Output X = HIGH (1) Continuously through the first half.

Second Half (input B is HIGH):

*Input B = HIGH(1) throughout this period

*Input A = toggles between HIGH and LOW

When input A goes HIGH(1): Both inputs are now HIGH,
So Output X = LOW(0)

When input A goes LOW(0): Both inputs aren't HIGH,
So Output X = HIGH(1)

Timing Diagram:

Input A : | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |

Input B : | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

Output X : | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 |

| <-- First Half --> | <-- Second Half --> |

9. Output waveform for 3 input NAND Gate

Rules for our current case:

- Output X is LOW only when all three inputs are HIGH at the same time.
- If any input is LOW, Output(X) is HIGH.

Interval 1(t1)

*Input A = HIGH(1), B = LOW(0), C= HIGH(1)

Since not all inputs are high, our Output(X) = HIGH(1)

Interval 2(t2)

*Input A = LOW(0), B = HIGH, C= LOW

Since A and C are LOW, our Output(X) = HIGH(1)

Interval 3(t3)

*Input A = HIGH, B = HIGH, C= HIGH

Since all inputs are HIGH, Output X = LOW(0)

Interval 4(t4)

*Input A = LOW, B = LOW, C= LOW

Since all inputs are LOW, Output X = HIGH(1)

Interval 5(t5)

*Input A = HIGH, B = HIGH, C= HIGH

Since all inputs are HIGH, Output X = LOW(0)

Interval 6(t6)

*Input A = LOW, B = HIGH, C= LOW

Since A and C are LOW, Output X = HIGH(1)

Timing Diagram for our 3-Input NAND Gate

Interval	t1	t2	t3	t4	t5	t6
Input A :	1	0	1	0	1	0
Input B :	0	1	1	0	1	1
Input C :	1	0	1	0	1	0

Output X :	1	1	0	1	0	1

10) Output waveform for 3 input NOR Gate

- Output is HIGH when all three inputs are LOW at the same time.
- If any input is HIGH, Output X is LOW.

Interval 1(t1):

Input A = HIGH, B = LOW, C = HIGH

So output X = LOW

Interval 2(t2):

Input A = LOW, B = HIGH, C = LOW

So output X = LOW(0)

Interval 3(t3):

Input A = HIGH, B = HIGH, C = HIGH

So output X = LOW(0)

Interval 4(t4):

Input A = LOW, B = LOW, C = LOW

So output X = HIGH(1)

Interval 5(t5):

Input A = HIGH, B = HIGH, C = HIGH

So output X = LOW(0)

Interval 6(t6):

Input A = LOW, B = HIGH, C = LOW

So output X = LOW(0)

Timing Diagram for our 3-Input NOR Gate

Interval :		t1		t2		t3		t4		t5		t6	
Input A :		1		0		1		0		1		0	
Input B :		0		1		1		0		1		1	
Input C :		1		0		1		0		1		0	

Output X :		0		0		0		1		0		0	

11.) Equivalence of NOR and Negative-AND gates

*NOR Gate is just an OR gate followed by a bubble.

For a high output just one of the inputs must be HIGH.

*Negative - AND gate is basically an AND gate with inverted outputs.

For a HIGH output, all inputs must be HIGH, if any input is HIGH, the output is LOW.

State 1: A = 0, B = 0

- NOR Gate: Neither input is HIGH, so Output(X) = HIGH(1)
- Negative - AND gate: both inputs are LOW, so Output (X) = HIGH(1)

State 2: A = 0, B = 1

- NOR Gate Output(X) = LOW(0)
- Negative -AND gate: Output(X) = (LOW)

State 3: A = 1, B = 0

- NOR Gate Output(X) = LOW(0)
- Negative -AND gate: Output(X) = (LOW)

State 4: A = 1, B = 1

- NOR Gate Output(X) = LOW(0)
- Negative -AND gate: Output(X) = (LOW)

Functional Proof Table (DeMorgan's Equivalent):

Input A	Input B	NOR Viewpoint	Negative-AND Viewpoint (Both 0 -> 1)	Output X
0	0	No HIGH inputs -> HIGH (1)	Both inputs LOW -> HIGH (1)	1
0	1	B is HIGH -> LOW (0)	Not both LOW -> LOW (0)	0
1	0	A is HIGH -> LOW (0)	Not both LOW -> LOW (0)	0
1	1	Both HIGH -> LOW (0)	Not both LOW -> LOW (0)	0

Both gates are equivalent based on De Morgan's theorem . When we look at the NOR gate, any high input gives us a low output, and when we look at the negative-AND gate, two low inputs are needed to produce a high output.

Demorgan's first law(NAND) = $(A * B) = A' + B'$

Demorgan's second law (NOR) = $(A+B)' = A' * B'$

12) Output waveform for Exclusive-OR(XOR) gate

Rules:

- Output is HIGH when two inputs are different.
- Output is LOW when two inputs are the same.

Input A is alternating between HIGH and LOW.

Input B is LOW during the first half and HIGH during the second half.

First half:

- Input B = LOW
- When input A is HIGH = (A = 1, B = 0)
Output(X) is HIGH(1)
- When input A is LOW = (A = 0, B = 0)
Output(X) is LOW(0)

Second half:

- Input B = HIGH
- When input A is HIGH = (A = 1, B = 1)
Output(X) is LOW(0)
- When input A is LOW = (A = 0, B = 1)
Output(X) is HIGH(1)

Timing Diagram for our 2-Input XOR Gate

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Input A : | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
Input B : | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
-----
Output X : | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
           |<Follows Input A>| < Inverted A > |
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*During the first half, our output follows input A because a HIGH on A creates different inputs(1 and 0)

*During the second half, our output X inverts input A because a HIGH on A results in identical inputs(1 and 1), forcing the output LOW.

13.) Output waveform for Exclusive-NOR(XNOR) gate

Rules:

- Output is HIGH when two inputs are the same .
- Output is LOW when two inputs are different.

Input A is alternating between HIGH and LOW.

Input B is LOW during the first half and HIGH during the second half.

First half:

- Input B = LOW
- When input A is HIGH = (A = 1, B = 0)
Output(X) is LOW(0)
- When input A is LOW = (A = 0, B = 0)
Output(X) is HIGH(1)

Second half:

- Input B = HIGH
- When input A is HIGH = (A = 1, B = 1)
Output(X) is HIGH(1)
- When input A is LOW = (A = 0, B = 1)
Output(X) is LOW(0)

Timing Diagram for a 2-Input XNOR Gate

Input A : | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |

Input B : | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |

Output X : | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

|< Inverted A > |< Follows Input A > |

